

University College Dublin
An Coláiste Ollscoile, Baile Átha Cliath

END OF MODULE EXAMINATION, 22 DEIREADH FÓMHAIR/OCTOBER 2026

COMP 10280

Programming I (Conversion)

Module Co-ordinator: Professor John Dunnion

Time allowed: 105 minutes.

Instructions for Candidates

Answer both questions.
Students should write programs in Python 3.
Pseudocode for algorithms should be included with programs.
Upload your work every 15 minutes.

Instructions for Invigilators

Students only have access to:

- Brightspace;
- Lecture slides for the module;
- Their own solutions to problems on Practical Sheets;
- The official Python Website, `http://www.python.org`; and
- A dictionary or translation Webpage (eg Google Translate).

No other Internet access is permitted.

1. A palindrome is a string that reads the same backwards as it does forwards. For the purposes of this exercise, letters, numeric digits (0–9) and the characters "+", "-", "*", "/" and "?" are considered, but all other characters are ignored. The case (uppercase/lowercase) of letters *is* to be considered. Also, the "?" character is a *wildcard character* that will match any character (including another "?"). Thus "abba", "A-9?-A" and "A2␣3b4b32A" are palindromes according to this definition, but "Abba", "abc9/cba", "a98a", "a-bba" and "Madam,␣I'm␣Adam!" are not (but "Madam,␣I'm␣adaM" is!).

   Write a program in Python 3 that prompts the user for a series of strings and checks whether each one is a palindrome according to the above definition. The program should continue until an empty string is entered. Your program should include a *non-recursive* function that checks whether a given string is a palindrome.

   Sample output from this program is as follows:

```
Enter a string (empty string to exit):  abba
abba is a palindrome.
Enter a string (empty string to exit):  A-9?-A
A-9?-A is a palindrome.
Enter a string (empty string to exit):  A2 3b4b32A
A2 3b4b32A is a palindrome.
Enter a string (empty string to exit):  Abba
Abba is not a palindrome.
Enter a string (empty string to exit):  abc9/cba
abc9/cba is not a palindrome.
Enter a string (empty string to exit):  a98a
a98a is not a palindrome.
Enter a string (empty string to exit):  a-bba
a-bba is not a palindrome.
Enter a string (empty string to exit):  Madam, I'm Adam
Madam, I'm Adam is not a palindrome.
Enter a string (empty string to exit):  Madam, I'm adaM
Madam, I' madaM is a palindrome.
Enter a string (empty string to exit):  x
x is a palindrome
Enter a string (empty string to exit):  xx
xx is a palindrome
Enter a string (empty string to exit):  x?
x?  is a palindrome
Enter a string (empty string to exit):  ?x
?x is a palindrome
Enter a string (empty string to exit):  ?xx
?xx is a palindrome
Enter a string (empty string to exit):
Finished!
```

[30 marks]

2. Write a program in Python 3 that prompts the user for a series of filenames and performs statistics on each of the files in turn.

   For each file, the program should calculate the total number of lines, characters, vowels, consonants, octal numeric digits, words and blank lines in the file. It should also calculate the maximum number of characters, vowels, consonants, octal numeric digits and words on a line in the file and print out the maximum and the line number on which each of these was found; if none of these was found, there should not be a message referring to the maximum. If the maximum number appears on more than one line, the first of these lines is printed.

   The program should also prompt the user for a string to search for in the file. If the user enters a string, the program should search for the string and either print out the line numbers where the string was found or print a message that the string cannot be found; if the user does not enter a string, a message should be printed out to that effect and the program should not search.

   When the program prompts the user for the name of a file that it should process, it should check that the file exists.

   The program should continue until an empty string is entered as a filename.

   Notes:

   - For the purposes of this exercise, *vowels* are the letters "A", "E", "I", "O", "U" and "Y" (in uppercase and lowercase).
   - For the purposes of this exercise, *consonants* are the letters (uppercase and lowercase) that are not vowels.
   - The octal numeric digits are 0, 1, 2, 3, 4, 5, 6 and 7.
   - For the purposes of this exercise, the newline character at the end of a line *is* to be counted as a character.
   - For the purposes of this exercise, a "word" is considered to be a string separated by arbitrary strings of whitespace characters (space, tab, newline, return and formfeed).
   - The program should check for the existence of each file in the current working directory/folder.
   - Four sample files, `file1.txt`, `file2.txt`, `file3.txt` and `file4.txt`, are available for download on the Brightspace site (thanks to Wikipedia and Cisco for the content!). Note that `file3.txt` is an empty file and `file4.txt` is a file with a single blank line.

Sample output from this program is as follows:

```
*****
Enter a file name: file1.txt
Enter a string to search for: string
Processing file file1.txt.
Statistics for file file1.txt:
Total number of characters: 642
Total number of vowels: 182
Total number of consonants: 227
Total number of octal digits: 56
Total number of words: 152
Total number of blank lines: 3
Total number of lines: 16

Maximum number of characters on a line: 97
This was (or the first occurrence of it was) on Line 5

Maximum number of vowels on a line: 30
This was (or the first occurrence of it was) on Line 11

Maximum number of consonants on a line: 53
This was (or the first occurrence of it was) on Line 5

Maximum number of octal digits on a line: 24
This was (or the first occurrence of it was) on Line 12

Maximum number of words on a line: 30
This was (or the first occurrence of it was) on Line 12

The string 'string' was not found.

*****
Enter a file name: file1.txt
Enter a string to search for: line
Processing file file1.txt.
Statistics for file file1.txt:
Total number of characters: 642
Total number of vowels: 182
Total number of consonants: 227
Total number of octal digits: 56
Total number of words: 152
Total number of blank lines: 3
Total number of lines: 16
```

```
Maximum number of characters on a line: 97
This was (or the first occurrence of it was) on Line 5

Maximum number of vowels on a line: 30
This was (or the first occurrence of it was) on Line 11

Maximum number of consonants on a line: 53
This was (or the first occurrence of it was) on Line 5

Maximum number of octal digits on a line: 24
This was (or the first occurrence of it was) on Line 12

Maximum number of words on a line: 30
This was (or the first occurrence of it was) on Line 12

The string 'line' was found in the following line(s):
5 7 8 15

*****
Enter a file name: file2.txt
Enter a string to search for:
Processing file file2.txt.
Statistics for file file2.txt:
Total number of characters: 5794
Total number of vowels: 1958
Total number of consonants: 2850
Total number of octal digits: 4
Total number of words: 862
Total number of blank lines: 0
Total number of lines: 38

Maximum number of characters on a line: 319
This was (or the first occurrence of it was) on Line 30

Maximum number of vowels on a line: 109
This was (or the first occurrence of it was) on Line 30

Maximum number of consonants on a line: 160
This was (or the first occurrence of it was) on Line 30

Maximum number of octal digits on a line: 3
This was (or the first occurrence of it was) on Line 24

Maximum number of words on a line: 47
This was (or the first occurrence of it was) on Line 20
```

```
No search string was supplied.

*****
Enter a file name: file3.txt
Enter a string to search for:
Processing file file3.txt.
Statistics for file file3.txt:
Total number of characters: 0
Total number of vowels: 0
Total number of consonants: 0
Total number of octal digits: 0
Total number of words: 0
Total number of blank lines: 0
Total number of lines: 0

Maximum number of characters on a line: 0

Maximum number of vowels on a line: 0

Maximum number of consonants on a line: 0

Maximum number of octal digits on a line: 0

Maximum number of words on a line: 0

No search string was supplied.

*****
Enter a file name: file4.txt
Enter a string to search for: abcdef
Processing file file4.txt.
Statistics for file file4.txt:
Total number of characters: 1
Total number of vowels: 0
Total number of consonants: 0
Total number of octal digits: 0
Total number of words: 0
Total number of blank lines: 1
Total number of lines: 1

Maximum number of characters on a line: 1
This was (or the first occurrence of it was) on Line 1

Maximum number of vowels on a line: 0
```

```
Maximum number of consonants on a line: 0

Maximum number of octal digits on a line: 0

Maximum number of words on a line: 0

The string 'abcdef' was not found.

*****
Enter a file name: file5.txt
Enter a string to search for:
File file5.txt does not exist.

*****
Enter a file name:
Finished!
```

[70 marks]

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